

C PROJECT REPORT ON

STRING UTILITY BOX

SUBMITTED TO

The Technical Department

KG College of Arts and Science

SUBMITTED BY

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Academic Year: **2025 – 2026**

Semester: **I**

DECLARATION

We hereby declare that the project titled “**String Utility Box**” submitted in partial fulfilment of the requirements for the course **BSC Computer Science** is a **record of our original work** carried out under the guidance of, Technical Trainer, Technical Department, KGiSL MicroCollege.

We further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

We have acknowledged all sources of information used in this project wherever applicable.

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Place: COIMBATORE

Date: _08/11/2025_

CERTIFICATE

This is to certify that the project entitled “**String Utility Box**” is a **bona fide work** carried out by the following students of **[Class/Batch]**, **[Department Name]**, **KG College of Arts & Science**, in partial fulfilment of the requirements for the course **[Course Name]**.

This project has been completed under my supervision and guidance during the academic period **[Month, Year]**.

It is further certified that this project is the **original work** of the students and has not been previously submitted for the award of any degree, diploma, or certificate at this or any other institution. **Project Team:**

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ABSTRACT

The **Attendance Management System** is a console-based project developed in the **C programming language** to efficiently record and manage student or employee attendance. The traditional manual attendance process is often time-consuming and prone to human errors. This system automates the process using file handling concepts in C, ensuring accuracy, simplicity, and quick access to attendance data.

The system allows users to **add student records**, **mark attendance**, and **generate attendance reports** through an easy-to-use menu-driven interface. Each student's data is securely stored in a binary file, which can be updated or retrieved as needed. The implementation demonstrates effective use of **structures**, **file operations**, and **modular programming** in C to create a real-world application. This project provides a practical example of how programming concepts can be applied to solve everyday administrative problems, particularly in educational or organizational environments. Future improvements could include integrating a graphical interface or database connectivity to enhance usability and scalability.

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1. INTRODUCTION

1.1 Objective

- To develop a user-friendly string processing tool
- To perform multiple string operations in one place
- To improve understanding of string manipulation concepts
- To reduce manual effort in text processing

1.2 Overview

This system allows users to:

- Strings are an essential part of programming used in almost every application. This project provides a collection of string operations such as reversing, counting characters, checking palindromes, converting case, and more. The system is designed to be user-friendly and efficient.

1.3 Features

- String reversal
- Palindrome check
- Uppercase & lowercase conversion
- Character count
- Word count
- Remove spaces
- Find substring
- Replace characters

2.PROGRAM ANALYSIS

2.1 Problem Definition

Handling multiple string operations separately can be time-consuming. There is a need for a single tool that performs various string operations in one place efficiently.

2.2 Project Overview

The String Utility Box is a menu-driven program where users can input a string and choose different operations to perform on it. The system processes the input and displays results instantly.

2.3 Scopes and Limitations

Scope:

- Useful for students and beginners
- Can be extended to advanced text processing
- Can be integrated into larger applications

Limitations:

- Handles only basic string operations
- Limited to small input sizes
- No advanced NLP features

3.SYSTEM DESIGN

3.1 Modules

- Input Module: Accepts string from user
- Processing Module: Performs selected operation
- Output Module: Displays results
- Control Module: Manages menu and user choices

4. METHODOLOGY

The development of the Attendance Management System project follows a structured methodology that includes five main stages: **Problem Analysis, System Design, Program Development, Testing & Validation, and Result & Documentation.**

4.1. Problem Analysis

Understanding user needs for common string operations and designing a system to combine them into a single tool.

4.2. System Design

The system is designed as a menu-driven program with modular functions for each operation.

4.3. Program Development

The program is developed using Python (or any language), implementing functions for each string operation.

4.4. Testing and Validation

- Tested with different string inputs
- Checked edge cases (empty string, special characters)
- Verified accuracy of outputs

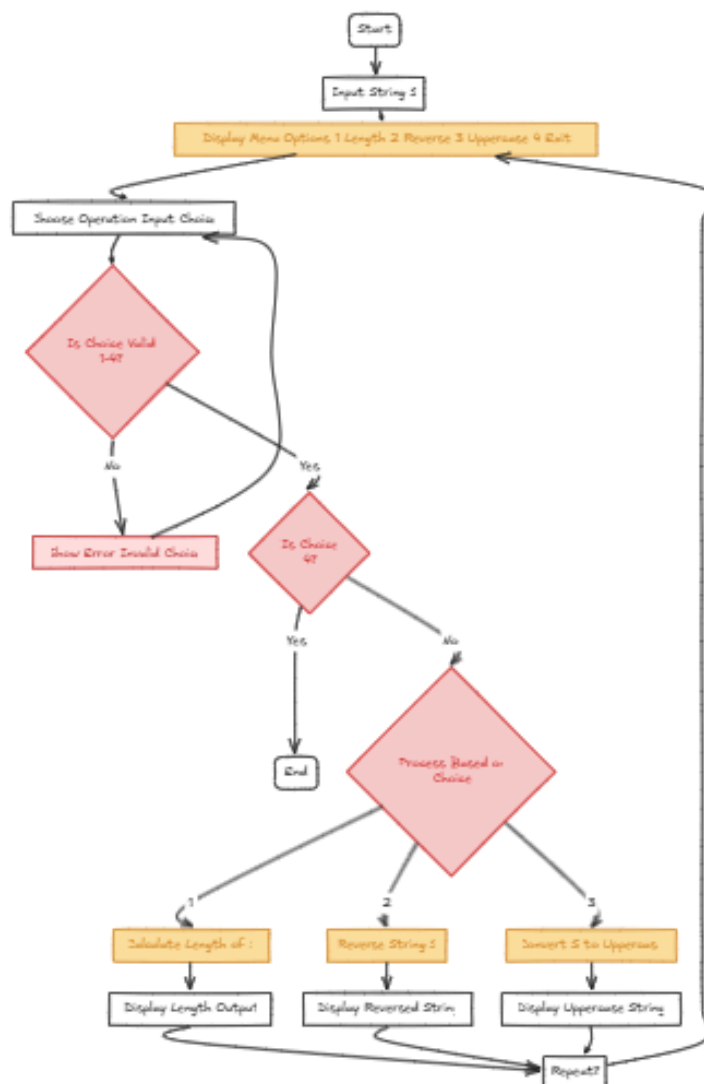
4.5. Result and Documentation

The program successfully performs all string operations and provides correct outputs.

4.6. Algorithm

1. Start
2. Input string from user
3. Display menu options
4. User selects operation
5. Perform selected operation
6. Display result
7. Repeat until exit
8. Stop

4.7. Flowchart



5.SOURCE CODE

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>
```

```
void reverseString(char str[]) {
    int len = strlen(str);
    printf("Reversed String: ");
    for(int i = len - 1; i >= 0; i--) {
        printf("%c", str[i]);
    }
    printf("\n");
}
```

```
void checkPalindrome(char str[]) {
    int len = strlen(str);
    int flag = 1;

    for(int i = 0; i < len / 2; i++) {
        if(str[i] != str[len - i - 1]) {
            flag = 0;
            break;
        }
    }
    if(flag)
        printf("Palindrome\n");
    else
        printf("Not Palindrome\n");
}
```

```
void countCharacters(char str[]) {
int vowels = 0, consonants = 0, digits = 0, spaces = 0;

    for(int i = 0; str[i] != '\0'; i++) {
        if(isdigit(str[i]))
            digits++;
        else if(str[i] == ' ')
            spaces++;
        else if(strchr("aeiouAEIOU", str[i]))
            vowels++;
        else if(isalpha(str[i]))
```

```
        consonants++;  
    }
```

```
printf("Vowels: %d\nConsonants: %d\nDigits:  
      %d\nSpaces: %d\n",  
      vowels, consonants, digits, spaces);  
    }
```

```
void toUpperCase(char str[]) {  
    for(int i = 0; str[i]; i++) {  
        str[i] = toupper(str[i]);  
    }  
    printf("Uppercase: %s\n", str);  
}
```

```
void toLowerCase(char str[]) {  
    for(int i = 0; str[i]; i++) {  
        str[i] = tolower(str[i]);  
    }  
    printf("Lowercase: %s\n", str);  
}
```

```
int main() {  
    char str[100];  
    int choice;
```

```
printf("Enter a string: ");fgets(str, sizeof(str), stdin);
```

```
    // Remove newline  
    str[strcspn(str, "\n")] = '\0';
```

```
    do {  
printf("\n--- STRING UTILITY BOX ---\n");  
        printf("1. Reverse String\n");  
        printf("2. Check Palindrome\n");  
        printf("3. Count Characters\n");  
        printf("4. Convert to Uppercase\n");  
        printf("5. Convert to Lowercase\n");  
        printf("6. Exit\n");
```

```
        switch(choice) {
            case 1:
                reverseString(str);
                break;
            case 2:
                checkPalindrome(str);
                break;
            case 3:
                countCharacters(str);
                break;
            case 4:
                toUpperCase(str);
                break;
            case 5:
                toLowerCase(str);
                break;
            case 6:
                printf("Exiting...\n");
                break;
            default:
                printf("Invalid choice!\n");
        } while(choice != 6);

        return 0;
    }
}
```

6. INPUT AND OUTPUT SCREENS

output:

```
Enter a string: madam

--- STRING UTILITY BOX ---
1. Reverse String
2. Check Palindrome
3. Count Characters
4. Convert to Uppercase
5. Convert to Lowercase
6. Exit
Enter your choice:
```

INPUT ONE:

```
Enter your choice: 1
Reversed String: madam

--- STRING UTILITY BOX ---
1. Reverse String
2. Check Palindrome
3. Count Characters
4. Convert to Uppercase
5. Convert to Lowercase
6. Exit
```

INPUT TWO:

```
Enter your choice: 2
Palindrome

--- STRING UTILITY BOX ---
1. Reverse String
2. Check Palindrome
3. Count Characters
4. Convert to Uppercase
5. Convert to Lowercase
6. Exit
```

INPUT THREE:

```
Enter your choice: 3
Vowels: 2
Consonants: 3
Digits: 0
Spaces: 0

--- STRING UTILITY BOX ---
1. Reverse String
2. Check Palindrome
3. Count Characters
4. Convert to Uppercase
5. Convert to Lowercase
6. Exit
```

INPUT FOUR:

```
Enter your choice: 4
Uppercase: MADAM

--- STRING UTILITY BOX ---
1. Reverse String
2. Check Palindrome
3. Count Characters
4. Convert to Uppercase
5. Convert to Lowercase
6. Exit
```

INPUT FIVE:

```
Enter your choice: 5
Lowercase: madam

--- STRING UTILITY BOX ---
1. Reverse String
2. Check Palindrome
3. Count Characters
4. Convert to Uppercase
5. Convert to Lowercase
6. Exit
```

INPUT SIX:

```
Enter your choice: 6
Exiting...
```

7.DISCUSSION & CONCLUSION

7.1 Discussion

The project demonstrates how multiple string operations can be integrated into a single system. It improves efficiency and reduces time for performing repetitive tasks.

7.2 Conclusion

The String Utility Box is a useful and simple tool for string manipulation. It is easy to use and can be enhanced further with more advanced features like pattern matching and text analysis.

8. REFERENCES

□ Balagurusamy, E. *Programming inANSIC*

- W3Schools (C Tutorial)
- GeeksforGeeks (C Programming)
- GCC Compiler